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Effect of defects in capacitance-voltage measurement of doping profiles in Ga_2O_3

J.V. Li [a,b,*], A.T. Neal [a], T.J. Asel [a], Y. Kim [a], S. Mou [a]

[a] Air Force Research Laboratory, Materials and Manufacturing Directorate, 2179 12th St, Wright-Patterson AFB, OH 45434, USA
[b] Azimuth Corporation, 2970 Presidential Dr. #200, Fairborn, OH 45324, USA

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ABSTRACT

We investigate the intermixing effects of doping profile and the carrier emission from deep levels in the capacitance-voltage measurement of the wide bandgap semiconductor material of β-Ga_2O_3. Specifically, we find that the spatial non-uniformity of doping measured under practical conditions is substantially contributed by artifacts due to carrier emission from deep levels. We develop a procedure to measure the hysteresis in cyclic capacitance-voltage experiments in dark and at room temperature for probing of the deep levels contributing to the apparent doping profile. Analysis of this hysteresis in the dynamic electrostatic framework of a Schottky junction containing deep levels enables more accurate determination of the doping density and its spatial distribution, and simultaneously the extraction of energy, density, and capture cross-section of the deep levels.

1. Introduction

β-Ga_2O_3 and other polymorphs of Ga_2O_3 [1–3] are promising ultra-wide-bandgap semiconductor materials under intensive investigation for future power and radio-frquency devices. Perhaps the most pressing issue at present for β-Ga_2O_3 is the continuous improvement of its electrical properties, among which doping and defects [4] are of particular importance because they affect multiple aspects of the technological development such as crystal growth, device fabrication, reliability, and radiation hardness. Doping and deep defects are also entangled in underlying physics and chemistry. Even their electrical characterization methods share the common bases, i.e., probing the charge response (e.g., capacitance) of the material under different conditions of DC bias voltage, frequency-time scale, and temperature. This work centers on one of the most widely used electrical measurement techniques for doping characterization—capacitance-voltage (CV) profiling—for the purpose of doping and defect investigation in β-Ga_2O_3.

Doping uniformity is one aspect crucial to the feasibility and yield of larger-scale manufacturing of the emergent β-Ga_2O_3-based electronics. This work is concerned with vertical doping uniformity, which might be caused by various mechanisms such as surface riding of dopant atoms (e.g., Sn and Ge) during epitaxial growth, out diffusion of compensating impurities (e.g., Fe and Mg) from a semi-insulating substrate, and accumulation of ambient impurities (e.g., Si) prior to epitaxial growth. Figure 1 shows one doping profile (symbols), which was measured by the capacitance-voltage technique using the common Mott–Schottky method [5] from a β-Ga_2O_3 substrate grown by the edge-defined film-fed growth method, exhibiting an increase in doping density of ~20% per 1 μm although in generally good agreement with Hall measurement [6]. The apparent increase of doping with depth is also reported by many other CV measurements on β-Ga_2O_3 [7–11] excluding those with known causes of non-uniformity such as delta-doping, ion implantation, and radiation damage. The ubiquity of the non-uniformity observation, its indifference of the growth and doping methods, and the large range of non-uniformity (e.g., from 15% to 140% over the depth of 100 s of nm in Fig. 1), is rather surprising and inexplicable from sheer growth point of view therefore warrants a close scrutiny.

CV hysteresis in Schottky and p-n junctions, which shares a common electrostatic basis underlying the threshold voltage hysteresis and instability in field-effect transistors, has been observed and thought to originate from the presence of deep states in the bandgap. Based on CV models by Goodman [5] and Sah et al. [12], Senechal and Basinski [13] experimentally extracted the density of the Au acceptor in n-Si. Glover [14] then extended the method by Senechal and Basinski [13] to measure the density and energy of a donor level 0.83 eV below the conduction band of n-GaAs. Nozaki and Milnes [15] provided a thorough theoretical foundation via a numerical analysis of the CV hysteresis due

* Corresponding author at: Air Force Research Laboratory, Materials and Manufacturing Directorate, 2179 12th St, Wright-Patterson AFB, OH 45434, USA.
E-mail address: jian.li.1.ctr@us.af.mil (J.V. Li).

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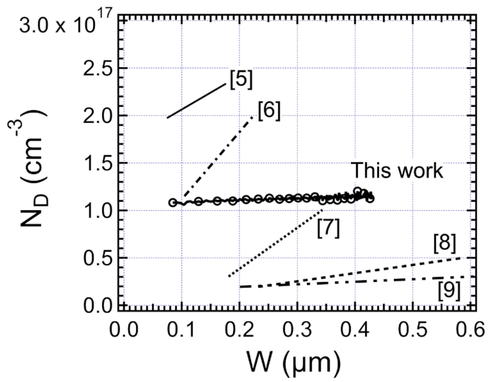

Fig. 1. The apparent doping profile obtained from CV scan in reverse direction (symbols) of a β-Ga_2O_3 Schottky diode exhibits non-uniformity. Published CV doping profiles [5–9] showing similar non-uniformity are sketched for reference.

to deep levels responding to a cyclic reverse bias. Balland et al. [16] further emphasized the difference of equilibrium and non-equilibrium CV scans as the cause of hysteresis and developed a method to extract the deep level density profile in phosphorous-implanted Si. More recently, Bailey et al. [17] developed the fast CV method to determine the doping profile in Cu(In,Ga)Se_2 with high density of deep levels. It is worth noting that Irmscher et al. [18] reported CV hysteresis in a β-Ga_2O_3 Schottky diode without accompanying analysis but the subject of the paper was on deep levels therefore hinting their implicit role in CV hysteresis.

The deep defect levels [4] in β-Ga_2O_3 have been detected and characterized by deep-level transient spectroscopy (DLTS) [18,19] and its variants such as deep-level optical spectroscopy (DLOS) [19] and admittance spectroscopy [6], which requires cryogenic temperature control, a vacuum sample environment, and sophisticated instrumentation for transient [20], frequency-sweeping [21], and optical spectroscopic measurements [19]. The lighted CV (LCV) technique [22] is built upon CV measurement to profile the density of deep levels and has found successful applications in wide-bandgap semiconductors in particular GaN [22]. However, the LCV technique requires precision control of illumination wavelengths and depends on prerequisite parameters such as deep-level energy that are extracted by additional DLOS experiment. This work explores the case of using illumination-less steady-state room-temperature CV technique for extracting the electrical properties of deep defect levels in β-Ga_2O_3.

We therefore investigate in this work the relationship between doping non-uniformity and carrier dynamics in deep levels in β-Ga_2O_3. We speculate that the carrier density non-uniformity is in part contributed by an artifact due to carrier emission from deep levels and seek experimental evidences from analytical electrostatic modeling of CV hysteresis. We aim to separate the contributions of doping and deep levels to more accurately quantify their respective parameters, i.e., doping density and spatial distribution for the former, and energy, density, and capture cross-section for the latter. This set of information is all measured and extracted at room temperature without sophisticated equipment.

2. Experiment

The ($\bar{2}$01) unintentionally doped n-type β-Ga_2O_3 substrates were commercially procured from Tamura Corporation, Japan, who grew the crystal using the edge-defined film-fed growth method. The 10 × 15 mm^2 substrates were received with one chemically-mechanically polished surface and one as-lapped surface. After dicing the substrate into 5 × 5 mm^2 pieces and going through solvent cleaning, we formed the ohmic contact by sputtering deposition of a 50 nm/1000 nm Ti/Au metal layer to the as-lapped side of the substrate. The samples then went through contract resistance improvement through annealing in a tube furnace with Ar gas flow using a 15-min ramp from room temperature to 450 °C. The Schottky contact was formed by sputtering deposition of a 150 nm Au metal layer on the polished side of the substrate. No intentional annealing was applied to the samples after the Schottky junction formation. Devices used in this work (Fig. 2) have circular Schottky contacts with active areas ranging from 0.0175 cm^2 to 0.003 cm^2. The electrical measurements were conducted on devices with and without wire bonding. All measurements were conducted in dark at room temperature (specifically, 17 °C) without using a cryostat or other temperature control equipment.

We used a Keithley 4200A-SCS parameter analyzer equipped with a 4210-CVU capacitance-voltage unit and an Agilent 4294A impedance analyzer for the CV measurements. The current-density vs. voltage characteristics measured at room temperature is shown in Fig. 2. The doping profile was calculated from the measured capacitance as the DC bias voltage was scanned monotonically in a range from +0.5 to −15.0 V. In this work, we fixed the following conditions for the CV measurement: $f = 100$ kHz, $V_{AC} = 35$ $mV_{peak\text{-}peak}$, and DC bias step size $V_{DC,step} =$ 250 mV. The 100 kHz frequency allows the shallow donor levels but not the deep levels to respond to the AC modulation. Note that the deep levels can still respond to the change of the DC bias voltage, which is key to the experiments and analysis conducted in this work. While keeping the above conditions fixed, we varied those conditions most directly relevant to the emission-capture dynamics of deep levels: the direction of V_{DC} scan, the V_{DC} scan time, and the time spent holding the device under constant bias voltage V_{DC} at either end of the scan range. For the interest of this work, the two instruments of Keithley 4200A-SCS and Agilent 4294A are functionally identical except for the fastest single point capacitance measurement time, which is 36 ms for the former and 3 ms with accuracy tradeoff for the latter.

A generic CV measurement cycle used is this work is illustrated in Fig. 3. Starting with the junction in a thermal equilibrium state, e.g., after being stored in dark and at zero bias for a long period of time (typically overnight), the CV cycle consists of four steps: 1) measure the capacitance transient $C(t)$ while holding the junction at a voltage V_F for a duration of t_F for the device to reach State F1; 2) conduct a reverse CV scan in which V_{DC} is changed from V_F to a reverse bias voltage V_R over scan time t_{FR} for the device to reach State R0; 3) measure the capacitance transient while holding the junction bias at V_R for t_R for the device to reach State R1; and 4) conduct a forward CV scan in which V_{DC} is changed from V_R to V_F over scan time t_{RF} for the device to reach State F0. V_F is either a moderate forward bias or zero. In the latter case, Step 1 for the first CV cycle is trivial because there is no capacitance transient. However, capacitance transients may still occur in Step 1 of the ensuing CV cycles because State F0 of the device may not be at equilibrium after Step 4. For reliable measurements, the device is held for sufficient long time t_F at V_F in order for $C(V_F, t_F)$ to reach the same steady-state value within experimental error, i.e., each CV cycle starts with the same equilibrium State F1. Such a requirement does not apply to $C(V_R, t_R)$ because the State R1 needs not be in equilibrium.

3. Results

As shown in Fig. 1, the doping profile measured by the reverse CV scan in Step 2 with the fastest scan rate allowed by Keithley 4200A-SCS (measurement time is 36 ms per bias point) exhibits a slight non-uniformity with the apparent doping density increasing with depth. After holding the bias at V_R for t_R in Step 3, a more strikingly non-uniform doping profile (Fig. 4) is produced by the forward CV scan in Step 4. The discrepancy between the apparent doping profiles measured by reverse and forward CV scans reveals a substantial hysteresis in the CV scan cycle. In metal-insulator-semiconductor capacitors, CV hysteresis is a routine tool [23] to detect and characterize the states at the insulator-semiconductor interface as well as in the bulk insulator or

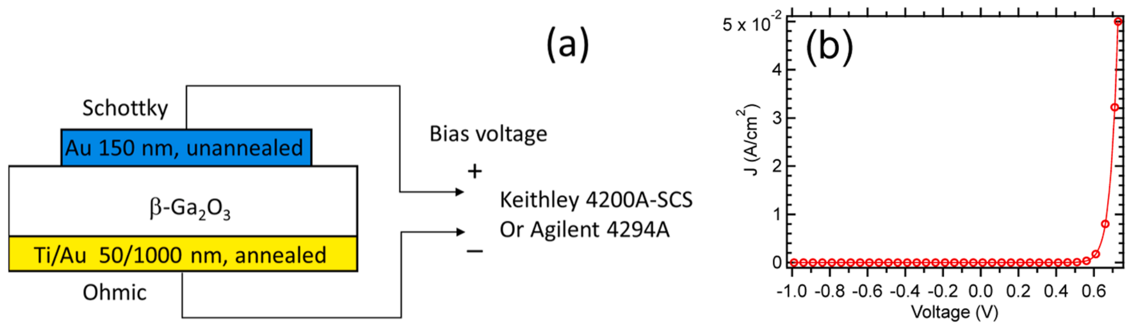

Fig. 2. (a) A diagram showing the devicestructure with measurement scheme; (b) the room-temperature current-density vs. voltage characteristics of a typical device.

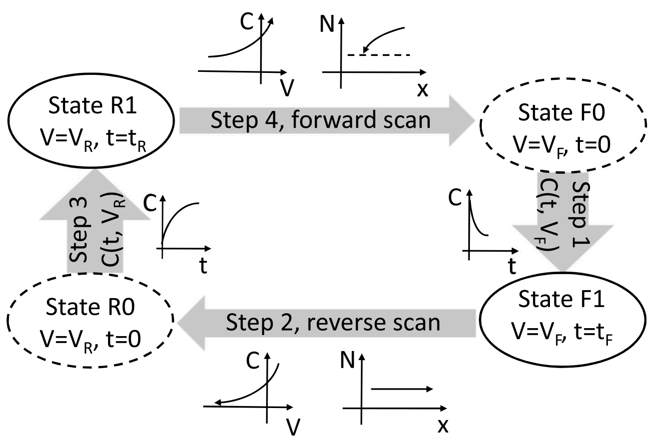

Fig. 3. Diagram of a generic CV measurement cycle used in this work.

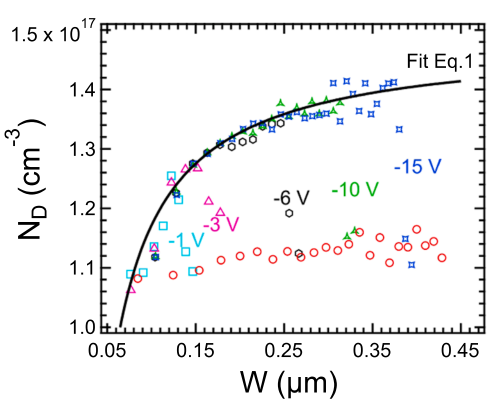

Fig. 4. Apparent electron density profile obtained by CV scanned in reverse (red, circles) and forward (all other colors and symbols) directions with various maximum reverse bias voltages. The solid curve is fitting of the forward CV scan starting from −15.0 V using Eq. (1).(For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

semiconductor. In the Schottky or *pn* junctions, the interface states [24] are of somewhat less concern but the deep levels in the bulk semiconductor predominantly contribute to CV hysteresis which is the focus of the following analysis [12–16].

First we make a few simplifications to obtain a qualitative understanding of the results in Fig. 1. Figure 5 shows the band diagrams of a semiconductor consisting of one shallow donor level responsible for uniform doping N_D and one deep level located at $E_C - E_T$, of uniform density N_T, and with electron emission time τ_{emit}. At V_F, the Fermi level E_F is above E_T so that all deep states capture electrons to reach the

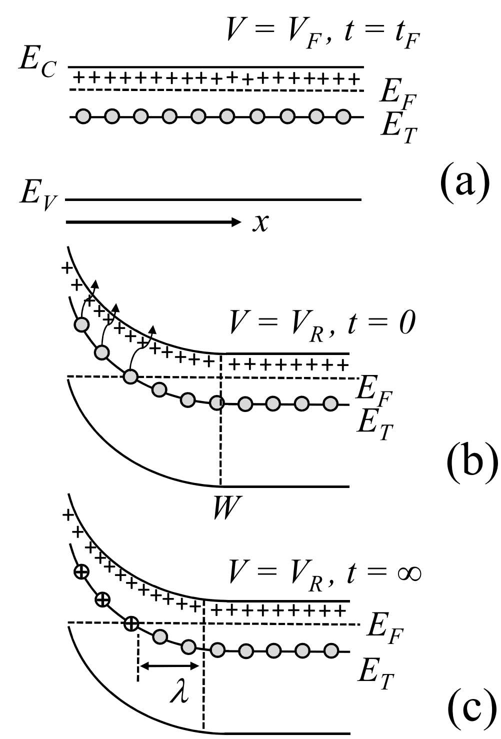

Fig. 5. Band diagrams of a simplified model of a semiconductor consisting of a Schottky junction one shallow donor level (crosses to symbolized their ionized state) and one deep level (circles) (a) filled with electrons with a bias at V_F, (b) in a non-equilibrium state immediately after the bias is switched to V_R, and (c) reaching equilibrium after holding at V_R for infinitely long.

neutral state. During the reverse CV scan in Step 2, the depletion region widens as a result of V_{DC} changing toward the reverse bias direction. Assuming a sufficiently fast V_{DC} scan from V_F to V_R, i.e., τ_{emit} much longer than the scan time t_{FR}, only the shallow donors respond to the change of V_{DC} with ionization and contribute to the positive space charge in this widened depletion region. The deep levels in the same region, though poised to be ionized because E_F is below E_t, do not have time to emit the electrons to contribute to the positive space charge and are thus in the non-equilibrium state. Hence, the carrier density extracted during the reverse CV scan is simply N_D. After holding the junction at V_R for a time longer than τ_{emit} during Step 3, all electrons are emitted from the previously occupied deep levels so they change from neutral to the 1^+ state. During the ensuing forward CV scan, the depletion region narrows. Assuming a sufficiently fast electron capture rate, both the positively charged deep levels and shallow donors in the depletion-region converted quasi-neutral region respond to the change of V_{DC} by returning to neutral equilibrium state. Consequently, the carrier density extracted during the forward CV scan of Step 4 is $N_D + N_T$.

The above analysis captures the essential physics but is too simplified to be of practical use beyond providing a rough estimate of N_T. The simplified model assumes all defects in the depletion region to be either neutral or positively charged, which overlooks the fact that E_F crosses E_t at a point some distance λ away from the depletion boundary W. Obviously, λ depends on the deep level energy and doping density at W [25], its effect being more prominent at bias voltages with $W \cong \lambda$, which is evident from the roll-off on the left side of the forward CV curve in Fig. 1. The right side of the forward CV curve approaches a plateau in qualitative agreement with the simplified model. Figure 4 shows that larger reverse biases reveal more complete plateau behavior even to the point of approximating the idea case, as a result of diminishing λ effect. Moreover, dynamic effects are inevitable as long as both the scan time and holding time are finite and comparable to the emission time constant τ_{emit}, which is not known a priori. In practice, there are also the needs to relax the assumption of one or both of N_D and N_T to be uniform. A more refined model is therefore required to separate the deep level contribution from the CV measurement in the presence of the above practical constraints and, in doing so, extract the full set of its electrical properties—energy, density, and capture cross-section.

Glover [13] recognized the deep levels as the source of slowly varying component of the capacitance during a reverse CV scan, which leads to time and depth dependent errors in the measured doping profile. To address this issue, Glover devised a non-monotonic CV scan scheme for the reverse CV scan where the capacitance at each bias point was measured only after the bias had been held steady for certain amount of time t. After the capacitance measurement at each bias point, V_{DC} always returned to V_F. In the concomitantly developed theoretical model, Grover solved the Poisson's equation in a Schottky junction with uniform doping and one deep level including full consideration of the λ effect. The time and depth dependent doping profile measured by the above per-point return-to-V_F reverse CV scan scheme [13] is:

$$N(W,t) = N_D(W) + u(W - -\lambda) \times N_T(W - -\lambda) \times (1 - -\lambda / W) \times [1 - -exp(- -t / \tau_{emit})], \quad (1)$$

where $u(x)$ is the step function and $\lambda \cong \ [2\varepsilon\ (E_F - E_T)\ /\ qN_D]^{1/2}$ is evaluated at $x = W$ [24]. Two extreme cases of Eq. (1) correspond to t equaling zero and infinity. For $t = 0$, there is no holding time at each V_{DC} point as it is scanned reversely, yielding $N(W, 0) = N_D(W)$ that agrees with the simplified analysis earlier. For $t = \infty$, the last term in Eq. (1) drops out, yielding an expression of $N(W, \infty)$ that takes into account the effects of N_T and λ, which is more accurate than the simplified analysis.

Because the CV scan scheme adopted by this work (Fig. 3) also return to V_F at the end of each CV scan cycle to reach the same equilibrium State F1, we inspect whether and how Eq. (1) is applicable to Steps 2 and 4 as well. It is immediately clear that the case of $t = 0$ in Eq. (1) describes the reverse CV scan in Step 2 if the scan time t_{FR} is much shorter than τ_{emit}. The case of $t \neq 0$ in Eq. (1) is less clear. For the extreme case of $t = \infty$, all electrons are emitted at the end of Step 3. Therefore, Eq. (1) with $t = \infty$ also describes the forward CV scan in Step 4 provided that electron capture by the deep levels is sufficiently fast. This is easily satisfied considering that the scan time is 5–6 s and the capturing rate $\sigma_n v_{th} n$ is on order of 1e−8 s^{-1} for most part of the neutral region newly converted from the previously depleted region under V_R. Therefore, Steps 2 & 4 of the CV scan cycle in Fig. 3 are describable by Eq. (1) while having the advantages of implementation ease and time efficiency over the per-point return-to-V_F reverse CV scan scheme. We substitute $t = \infty$ in Eq. (1) and use a single set of uniform doping and deep level parameters to fit the forward CV scan curves obtained at various V_R in Fig. 5. The fitting is reasonably good, especially for the section of curves where W is greater than λ, which enables the extraction of $N_D = 1.10\mathrm{e}17\ \mathrm{cm}^{-3}$, $N_T = 3.83\mathrm{e}16\ \mathrm{cm}^{-3}$ and $\lambda = 81.4$ nm. The fitting of Eq. (1) to regions where W is comparable with or smaller than λ is not as good due to second-order term of $d\lambda/dW$ [12]. Lastly, the extracted value of λ corresponds well to the intercepting point between the reverse and forward CV profiles, from which $\lambda = 78.0$ nm is extracted. The deep level energy of $E_T = 0.834$ and 0.809 eV calculated from λ extracted by fitting to Eq. (1) and identifying the intercepting point, respectively, agree well with previous DLTS measurements [17,18,26].

The preceding analysis satisfactorily separates the deep level contribution to CV scan from that by the shallow doping, extracts N_T, E_T, and determines N_D with better accuracy. However, since only the two extremes of $t = 0$ and ∞ are used, it is essentially a steady-state analysis assuming no time dependence of the charge state of N_T. As a result, the kinetic nature of the electron emission process is lost. Significantly, the critical parameter of capture cross section cannot be extracted from the above static analysis. Next, we fix the reverse bias voltage (−6.0 V) and conduct the forward CV scan (Step 4) at various holding time (Step 3) following the reverse CV scan. As shown in Fig. 6, the hysteresis increases with holding time as expected, eventually stabilizes at $t > 600$ s. This is a clear manifestation of the emission time of electrons from the deep level. The case of finite t in Eq. (1) is approximately applicable to the forward CV scan in Step 4 following a hold of the junction bias at V_R for t during Step 3 as long as the reverse and the forward scans (Steps 2 & 4) are much shorter than t. Using Eq. (1) to obtain the best fit to experimental CV forward scans taken at various hold times, one extracts $N_T(t)$ that varies with t. The time dependence of $N_T(t)$ is well described by $N_{T0}\ [1 - \exp(-t/\tau_{\mathrm{emit}})]$. Fitting this expression to the $N_T(t)$ (Fig. 6, inset), we further extract $\tau_{\mathrm{emit}} = 262$ s. From $\tau_{\mathrm{emit}}^{-1} = N_C\, v_{th}\, \sigma_n \exp(-E_T/k_BT)$, we obtain $\sigma_n = 3.49\mathrm{e}{-15}\ \mathrm{cm}^2$, which is in good agreement with previous reports using DLTS [17,18,26].

The analysis above demonstrates that the CV hysteresis and to some extent the non-uniformity are both in part caused by the presence of deep levels. After separating the defect contribution from free-carrier contribution, it is further possible to extract the full set of defect electrical properties. However, a uniform carrier density profile and a uniform defect density are both strong assumptions which are not always satisfied. In order to relax these two approximations, we look beyond the hysteresis of the N-vs-W curves and instead analyze its root cause of the hysteresis of C-vs-V curves. The reverse CV scan measures the non-equilibrium capacitance because the deep levels in the depletion region did not have time to emit electrons to reach their equilibrium states. The forward CV scan after holding the junction at V_R for a time longer than τ_{emit} measures the equilibrium capacitance because the deep levels have reached their equilibrium states. As shown in Fig. 7, the voltage it takes to reach certain capacitance value under the non-equilibrium condition is therefore different from that required to reach the same capacitance at equilibrium. The difference of these two voltages, ΔV, is a measure of the extent of hysteresis and reflects the necessary change of bias in order to ionize deep levels from 0 to $W_\lambda \equiv W$

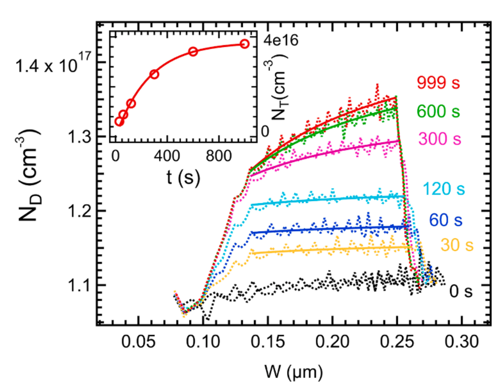

Fig. 6. Apparent doping profile obtained by forward CV scan following various holding time at the maximum reverse bias voltage of −6.0 V. Inset: fitting N_T vs. holding time to $N_{T0}\ [1 - \exp(-t/\tau_{\mathrm{emit}}]$ to extract emission time constant $\tau_{emit} = 262$ s.

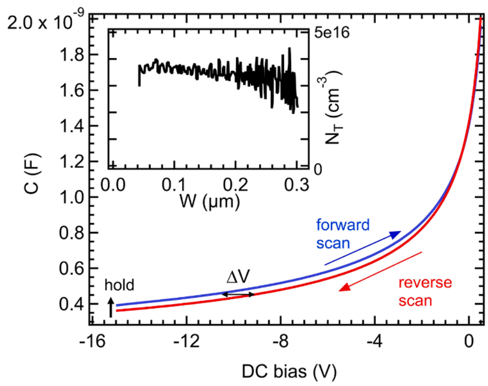

Fig. 7. CV hysteresis caused by electron emission from the deep levels is observed in a cycle of reverse scan (red), holding the bias at V_R, and then forward scan (red). Inset: the depth profile of the deep level N_T is extracted using Eq. (3).(For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$- \lambda$.

Balland et al. [15] developed a model that include both the defect and free-carrier contributions where neither N_T nor N_D is uniform with respect to x. A key result of the model is that N_T from 0 to W_λ is the only contributor to ΔV, i.e.,

$$\Delta V = (q / \varepsilon) \int_0^{W-\lambda} N_T(x) x dx \tag{2}$$

Therefore, $N_T(x)$ can be solved from $\Delta V(W_\lambda)$ as:

$$N_T(W_\lambda) = (\varepsilon / q) W_\lambda^{-1} d(\Delta V) / dW_\lambda, \tag{3}$$

where W is obtained from the non-equilibrium capacitance, from which $N_D(W) = (\varepsilon/q)W^{-1}$ dV/dW is also calculated. As shown in Fig. 7, N_T shows a slight non-uniformity although the data points on both end of the profile were susceptible of excessive numerical differentiation errors subject to improvement in future works.

4. Discussion

The above analysis demonstrates that the doping non-uniformity seen in forward CV scan is largely caused by the presence of deep levels. In Fig. 1, however, there is still a conspicuous slope in the reverse CV scanned apparent doping profile: the doping density at the end of the reverse CV scan (V_R = –15.0 V, scan time 5.6 s) is ~7% higher than that at the beginning (+0.5 V). As shown in Fig. 8, the doping profile slope from reverse CV scan increases substantially as the V_{DC} scan time t_{FR} increases above 6 s. For t_{FR} less than 2 s, the doping profiles from reverse CV scan are not distinguishable within experimental errors. This non-uniformity is clearly attributable to partial emission of electrons from the 0.8 eV deep levels during the finite reverse CV scanning time. The shape of the doping profile obtained by a reverse CV is different from that obtained by a forward CV scan taken after the same holding time t_R at V_R (Fig. 6). The reverse CV scan with time t_{FR} evenly distributed across all voltage points from V_F to V_R results in progressively less electron emission from the deep levels as they approach $x = W$ at V_R. CV scans in Fig. 8, even with very long scan time, are therefore not an equilibrium scans suitable for analysis described earlier.

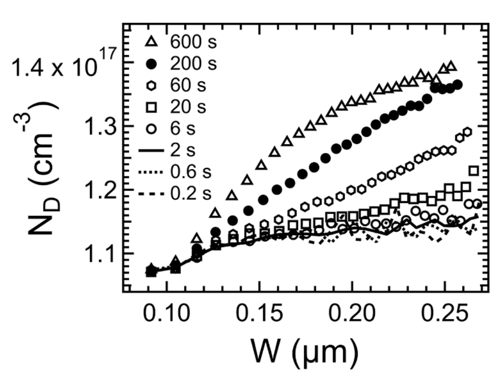

Fig. 8. Reverse CV scans with varying DC voltage scan time t_{FR}.

Because the emission time for the 0.8 eV deep level (~262 s) is much longer than the sweep time of 5.6 s, only a small portion of the electrons in this level contribute to the apparent carrier density during a reverse CV scan. Additionally, doping profile slope seen in the reverse CV scan in Fig. 1 may also be attributed in part to emission of electrons from the deep levels located at ~0.6 eV below E_C [17,18,26], whose emission time constant is ~ 10 s at room temperature and comparable to the total V_{DC} scan time of 5.6 s from +0.5 to –15.0 V. As far as the 0.6 eV level is concerned, the reverse CV scan conducted under this scan rate may not be strictly considered as being based on non-equilibrium capacitance measurement. As a result, a substantial portion of this deep level (density ~4.3e15 cm^{-3}) is included in the apparently measured N_D. Using scan time t_{FR} of 5.6 s as the approximated emission time, one can apply the same time dependent analysis used in Fig. 6 to estimate the combined contribution of electron emission from 0.6 to 0.8 eV deep levels as ~5e15 cm^{-3}, which nearly accounts for the doping non-uniformity in Fig. 1 and reduces it to below 2% after this correction.

It is known from previous measurements that there is at least one more deep level located at 1.0 eV [17,18,26] below the conduction band edge. The effect of the deeper 1.0 eV level is not significant under the experimental conditions in this work because of its exceedingly long emission time constant ~1e4 s [26]. In principle, as long as the emission time constant on one deep level τ_{emit} is clearly separated from those of the other deep levels τ_{other}, it is reasonable to expect the observation of CV hysteresis between a reverse scan ($\tau_{emit} > t_{FR} > \tau_{other}$) and a forward scan following a long hold $t_R > \tau_{emit}$ at V_R. A corollary follows that the most reliable CV measurement of doping profile is a reverse scan conducted with the highest AC modulation frequency f and shortest DC scan time t_{FR} afforded by the measurement system. Quantitative extraction of the properties of multiple deep levels may be obtained by apply the experiment and analysis described in this work iteratively. The detection sensitivity of deep level density N_D/N_T is ultimately determined by the instrumentation error, which was expected to be ~1e – 5 [15]. When the emission time constant and the corresponding holding time become impractically long, additional means such as raising the temperature and using light illumination can be used to expedite the carrier emission process at V_R.

5. Conclusions

This work demonstrates that the CV measurement of doping profile is intricately mixed with the carrier exchange behavior of deep levels in β-Ga_2O_3 and, by extension, wide bandgap semiconductors in general. Specifically, the spatial non-uniformity of doping measured under commonly used conditions is significantly contributed by an artifact due to carrier emission from deep levels. CV hysteresis observed in cyclic scanning enables experimental detection of deep level interference and quantitative extraction of the electrical properties of doping and deep levels. Employing electrostatic modeling of a Schottky junction containing deep levels, we separate the contributions of doping and defects and simultaneously quantify their respective parameters, i.e., doping density and spatial distribution for the former, and energy, density, and capture cross-section for the latter. The experimental and analytical results are based on practical CV measurements at room temperature without uses of DLTS, light, or a cryostat.

CRediT authorship contribution statement

J.V. Li: Conceptualization, Methodology, Investigation, Writing – original draft. **A.T. Neal:** Investigation, Validation, Writing – review & editing. **T.J. Asel:** Investigation, Validation, Writing – review & editing. **Y. Kim:** Investigation, Validation. **S. Mou:** Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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